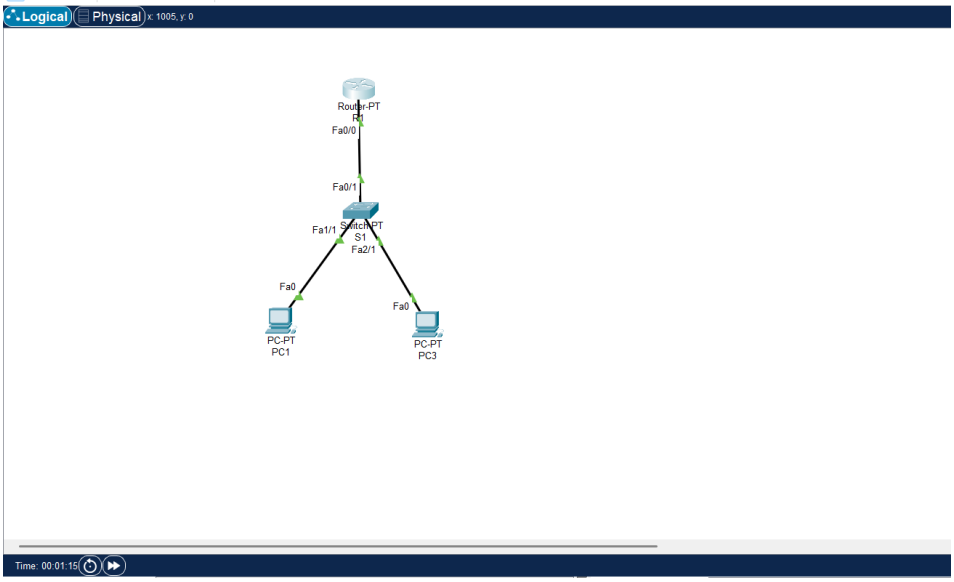
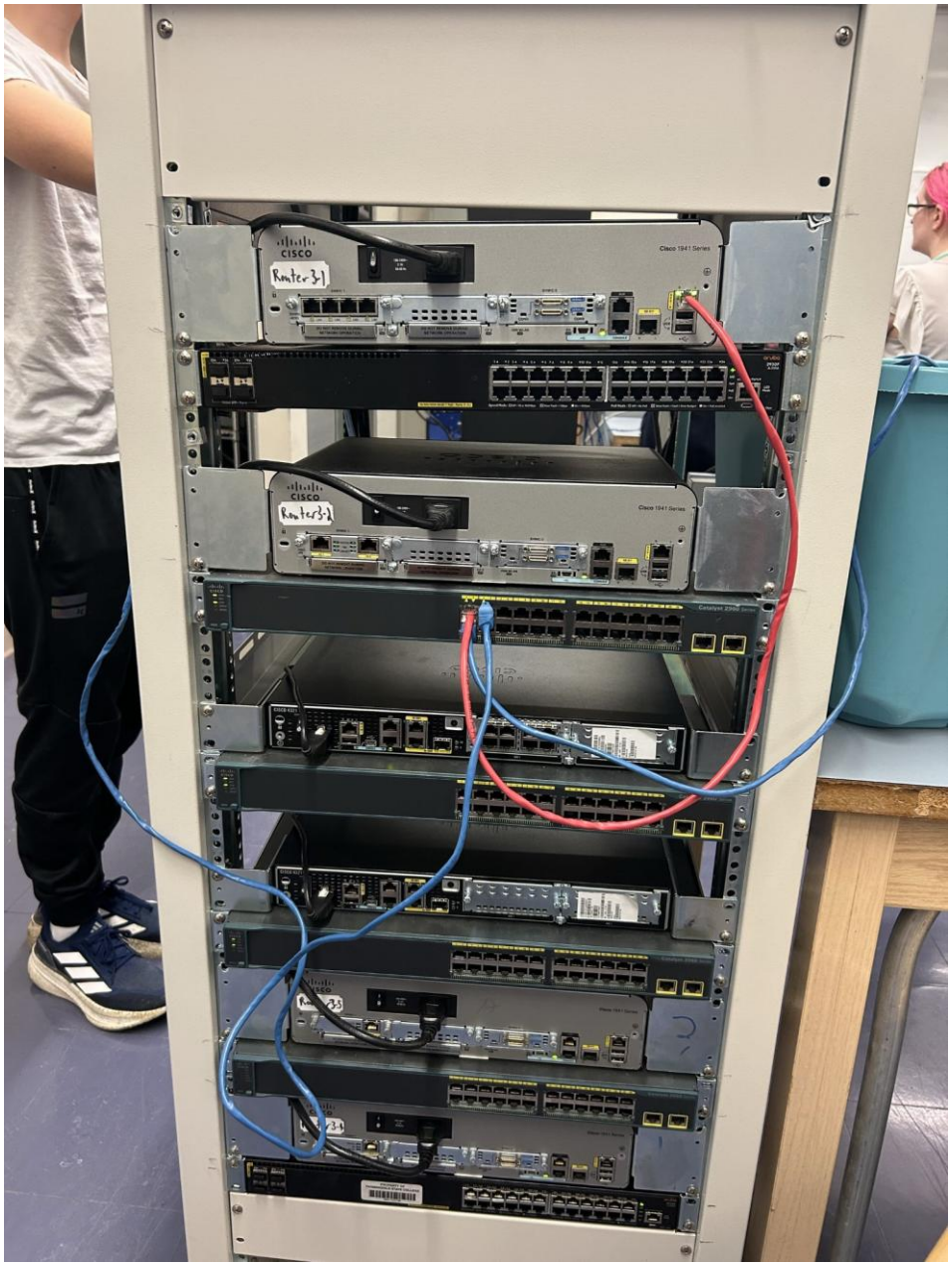
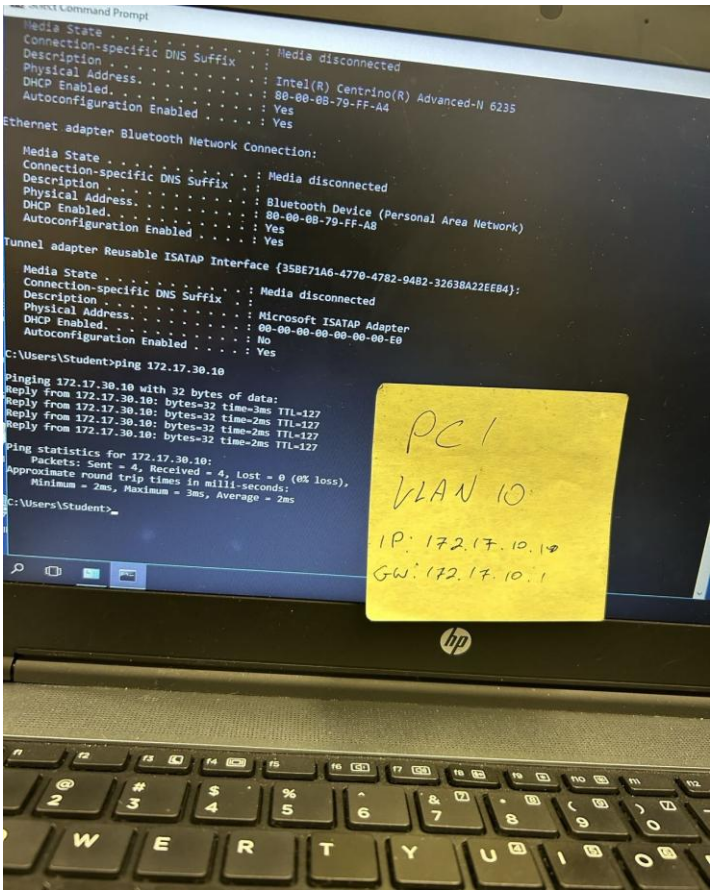
Lab 2

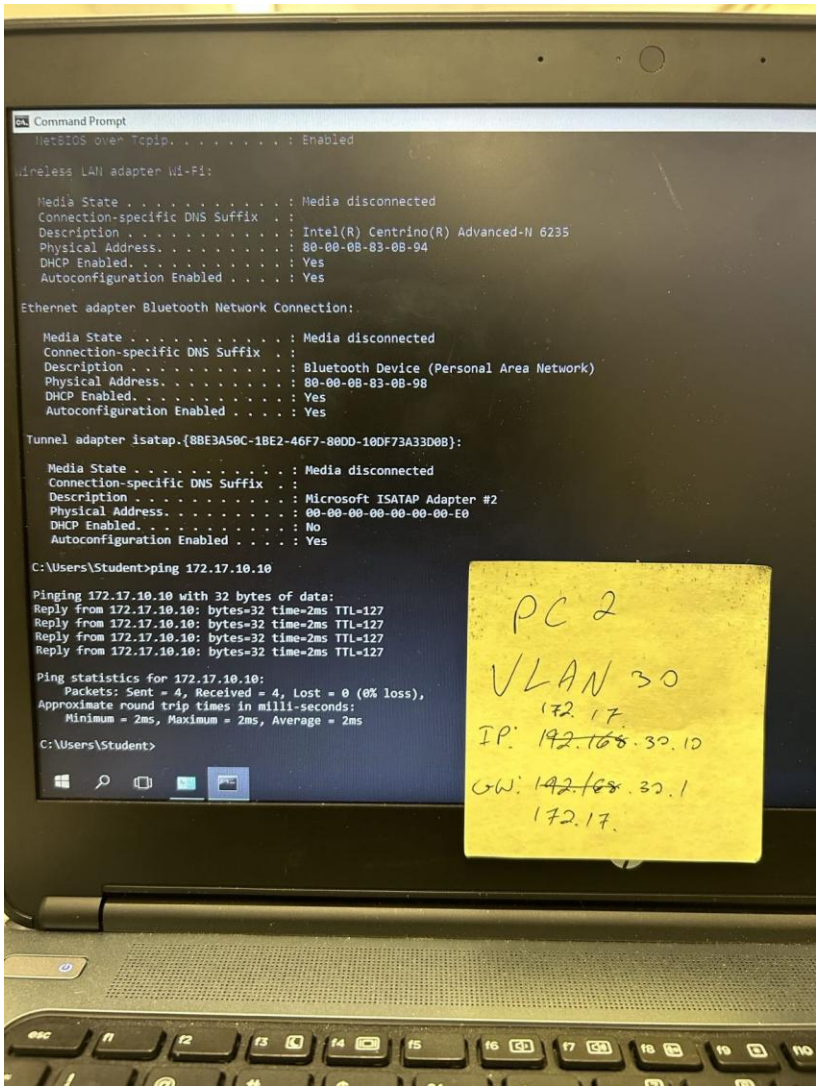
**Commented [1]:** Cisco Pkt: Vlan_Tagging

In the physical configuration, everything is the same as the Cisco packet tracer topology. The only difference is that the firewalls were disabled on the PCs in the physical setup due to it blocking packets, causing the ping commands to fail.

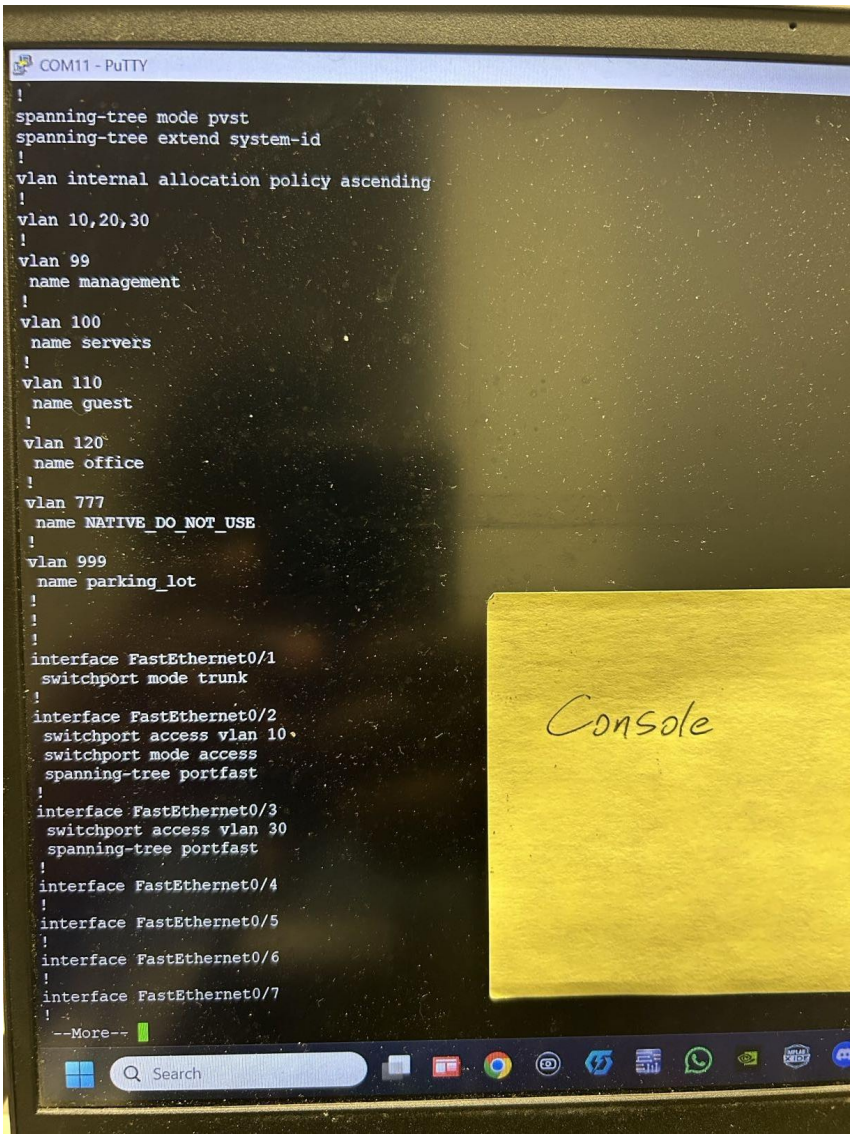



CISCO
Router3-1
Cisco 1941 Series
CISCO
Router3-2
Cisco 1941 Series

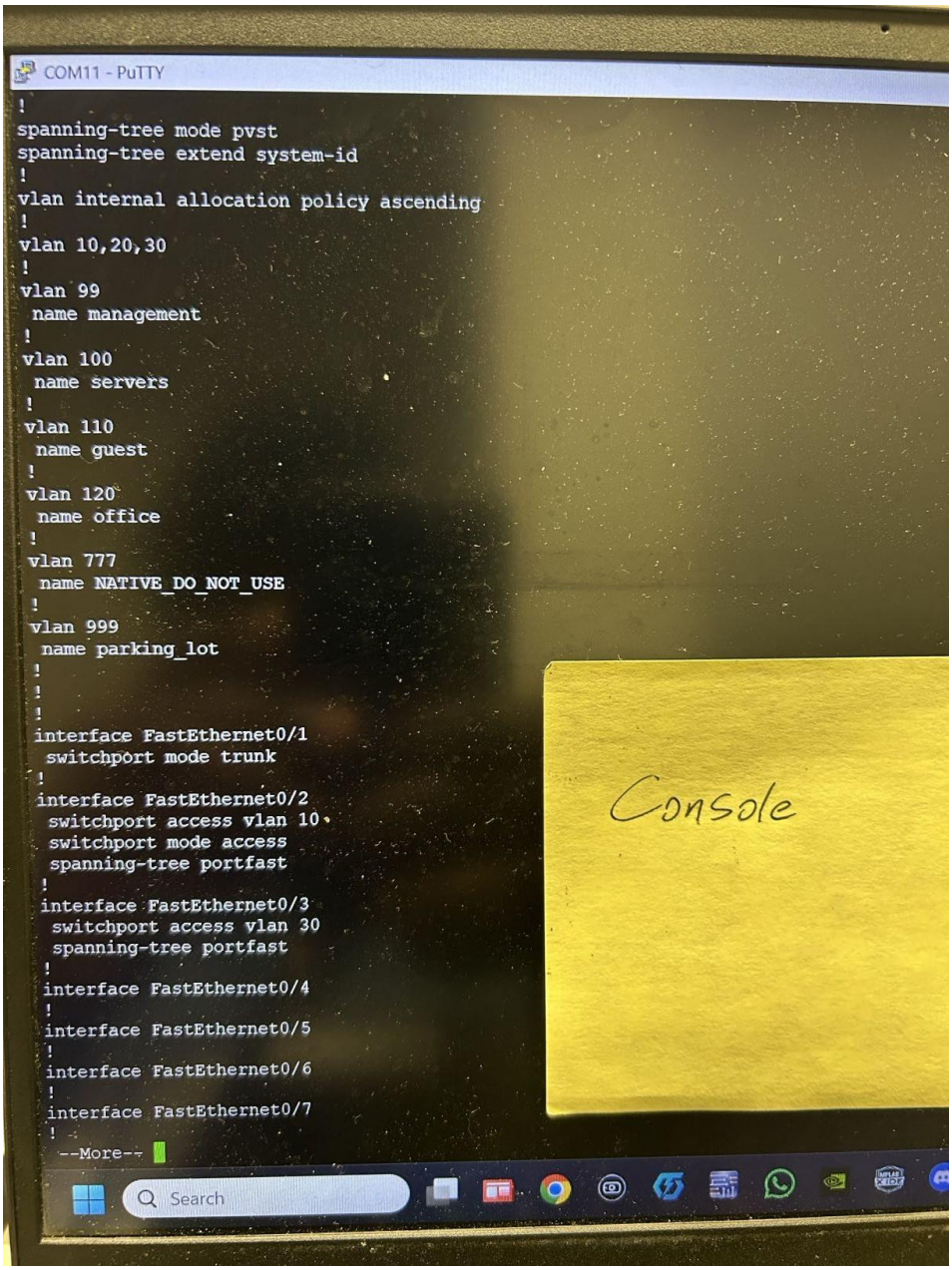

Command Prompt
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Intel(R) Centrino(R) Advanced-N 6235
Physical Address. . . . . : 80-00-0B-79-FF-A4
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Ethernet adapter Bluetooth Network Connection:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Bluetooth Device (Personal Area Network)
Physical Address. . . . . : 80-00-0B-79-FF-A8
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Tunnel adapter Reusable ISATAP Interface {35BE71A6-4770-4782-94B2-32638A22EEB4}:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft ISATAP Adapter
Physical Address. . . . . : 00-00-00-00-00-00-00-E0
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
C:\Users\Student>ping 172.17.30.10
Pinging 172.17.30.10 with 32 bytes of data:
Reply from 172.17.30.10: bytes=32 time=3ms TTL=127
Reply from 172.17.30.10: bytes=32 time=2ms TTL=127
Reply from 172.17.30.10: bytes=32 time=2ms TTL=127
Reply from 172.17.30.10: bytes=32 time=2ms TTL=127
Ping statistics for 172.17.30.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 2ms, Maximum = 3ms, Average = 2ms
C:\Users\Student>
PC1
VLAN 10
IP: 172.17.10.10
GW: 172.17.10.1


Command Prompt
NetBIOS over Tcpip. . . . . : Enabled
Wireless LAN adapter Wi-Fi:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Intel(R) Centrino(R) Advanced-N 6235
Physical Address. . . . . : 80-00-0B-83-0B-94
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Ethernet adapter Bluetooth Network Connection:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Bluetooth Device (Personal Area Network)
Physical Address. . . . . : 80-00-0B-83-0B-98
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Tunnel adapter isatap.{8BE3A50C-1BE2-46F7-80DD-10DF73A33D0B}:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft ISATAP Adapter #2
Physical Address. . . . . : 00-00-00-00-00-00-00-E0
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
C:\Users\Student>ping 172.17.10.10
Pinging 172.17.10.10 with 32 bytes of data:
Reply from 172.17.10.10: bytes=32 time=2ms TTL=127
Reply from 172.17.10.10: bytes=32 time=2ms TTL=127
Reply from 172.17.10.10: bytes=32 time=2ms TTL=127
Reply from 172.17.10.10: bytes=32 time=2ms TTL=127
Ping statistics for 172.17.10.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 2ms, Maximum = 2ms, Average = 2ms
C:\Users\Student>
PC 2
VLAN 30
IP: ~~192.168~~ 172.17.30.10
GW: ~~192.168~~ 172.17.30.1

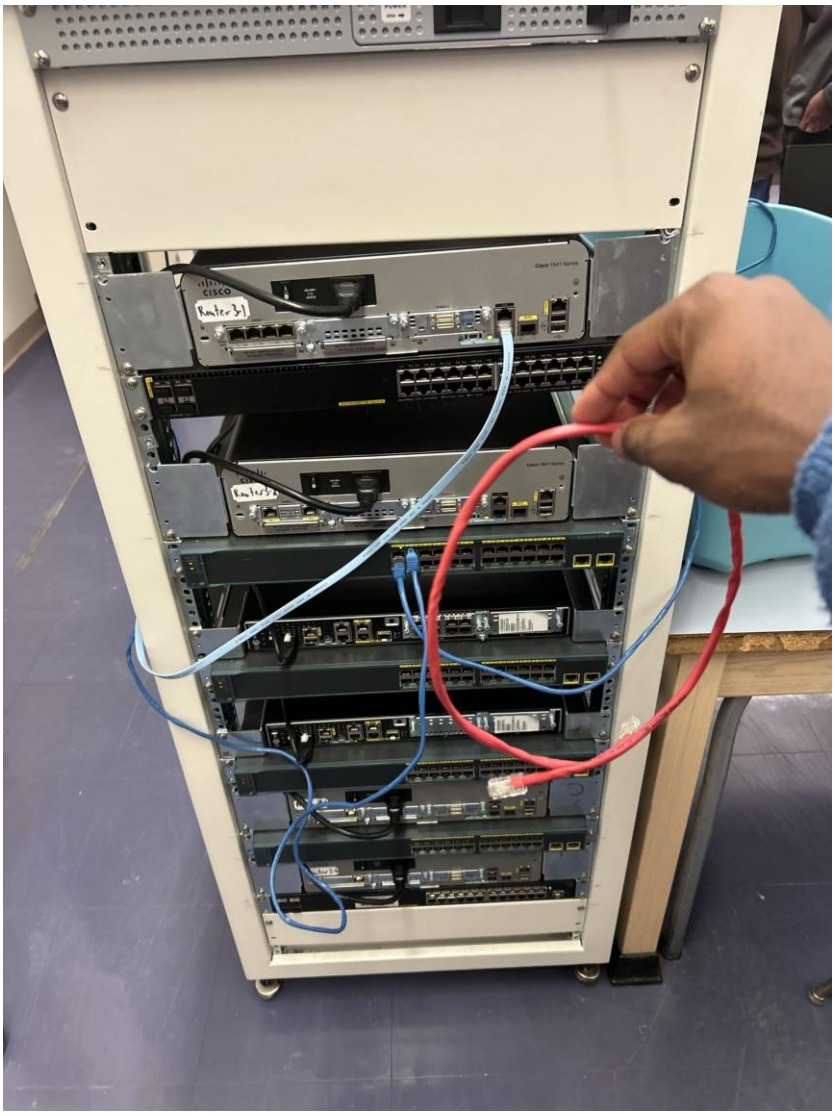
Switch Console:



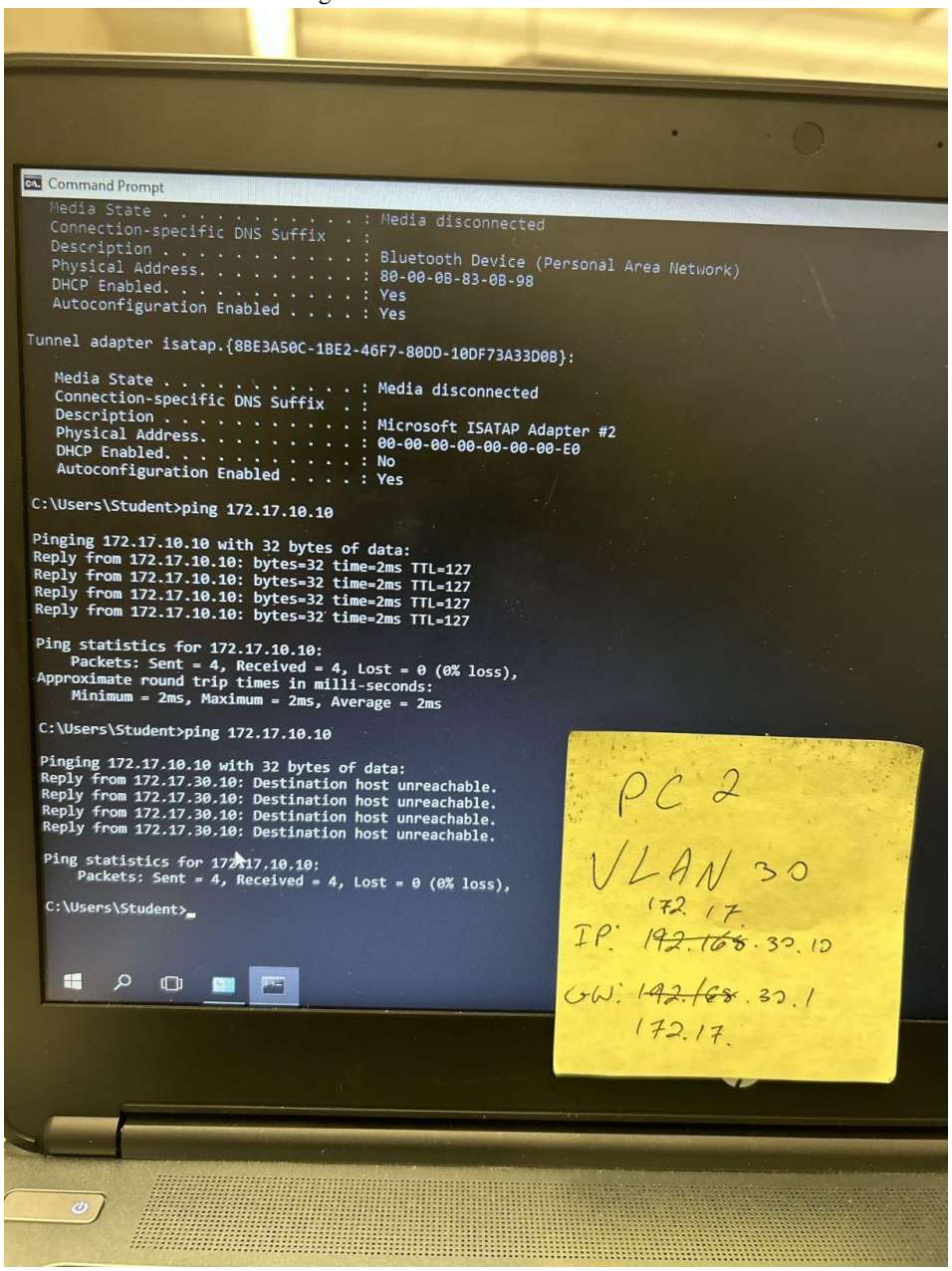
Router Console:



Disconnected Trunk:



Disconnected Trunk / Failed Ping:



**Router 3-1 Config:**

```
Router>en
Router#show run
Building configuration...
Current configuration : 902 bytes
!
version 12.2
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router
!
!
!
!
!
!
!
!
ip cef
no ipv6 cef
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
```

```
interface FastEthernet0/0
no ip address
duplex auto
speed auto
!
interface FastEthernet0/0.10
encapsulation dot1Q 10
ip address 172.17.10.1 255.255.255.0
!
interface FastEthernet0/0.30
encapsulation dot1Q 30
ip address 172.17.30.1 255.255.255.0
!
interface FastEthernet1/0
no ip address
duplex auto
speed auto
shutdown
!
interface Serial2/0
no ip address
clock rate 2000000
shutdown
!
interface Serial3/0
no ip address
clock rate 2000000
shutdown
!
interface FastEthernet4/0
no ip address
shutdown
!
interface FastEthernet5/0
no ip address
shutdown
!
ip classless
!
ip flow-export version 9
```

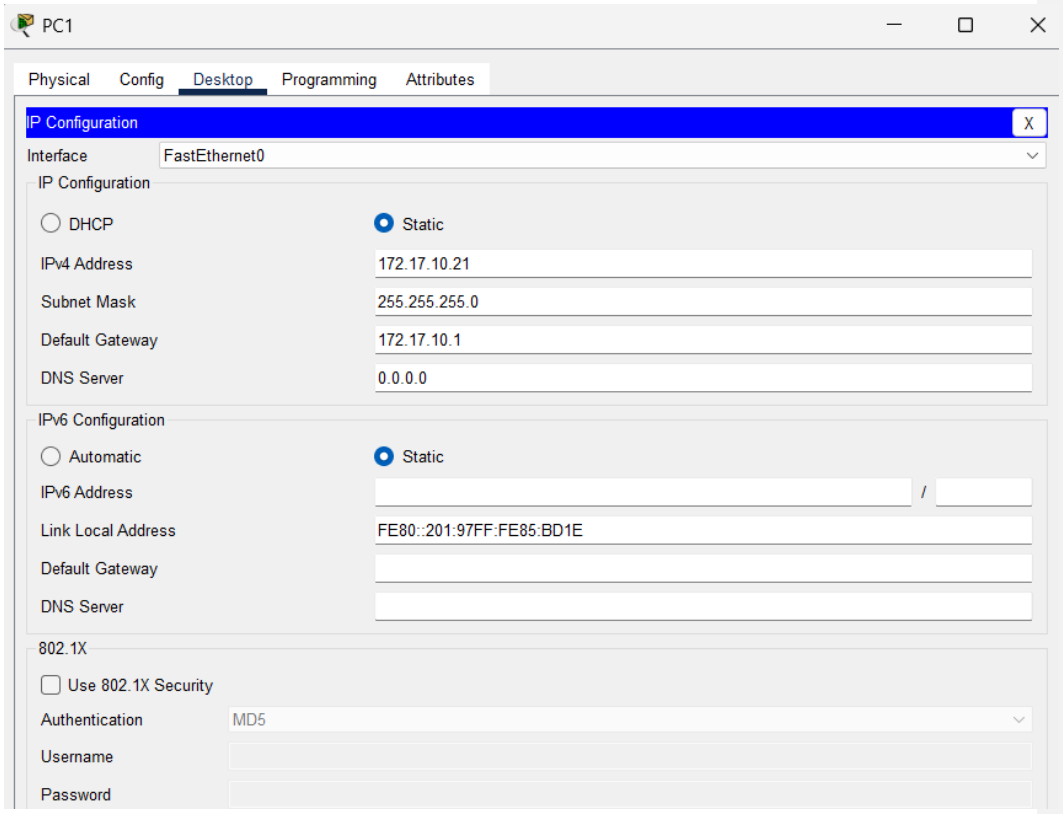
```
!
!
!
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
login
!
!
!
end
```

**<u>Switch (S1)</u>:**

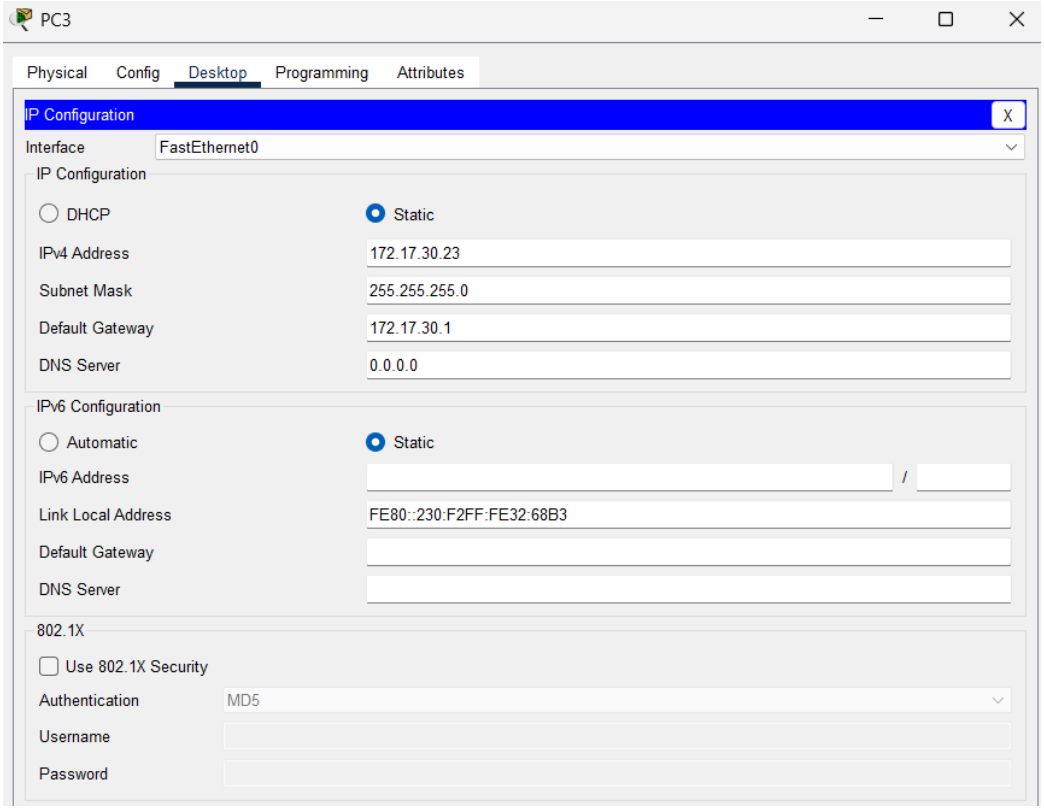
```
Switch>en
Switch#show run
Building configuration...
Current configuration : 624 bytes
!
version 12.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Switch
!
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
switchport mode trunk
!
interface FastEthernet1/1
switchport access vlan 10
switchport mode access
!
interface FastEthernet2/1
switchport access vlan 30
switchport mode access
!
interface FastEthernet3/1
!
interface FastEthernet4/1
!
interface FastEthernet5/1
!
interface Vlan1
no ip address
```

```
shutdown
!
!
!
!
line con 0
!
line vty 0 4
login
line vty 5 15
login
!
!
!
!
end
```

**PC1 Config:**


PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
172.17.10.21
Subnet Mask
255.255.255.0
Default Gateway
172.17.10.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::201:97FF:FE85:BD1E
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password

**PC3 Config:**


PC3
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
172.17.30.23
Subnet Mask
255.255.255.0
Default Gateway
172.17.30.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::230:F2FF:FE32:68B3
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password